

The Qubit: Physics, Phases and Hardware

Lecture Notes

In the previous lecture we took computation apart. We found the atoms of classical computation in the Boolean logic gates, rebuilt a half adder out of CNOT and Toffoli gates, and then discovered, by running two Hadamard gates in a row, that a quantum circuit does *not* behave like a probabilistic classical one. Tracking probabilities gave the wrong answer; tracking *amplitudes*, which may be negative and which therefore cancel, gave the right one. That was our first interference effect.

This lecture asks the two questions that follow immediately. First: what is the smallest piece of mathematics that accounts for the experiments, and why is it forced on us rather than chosen? Second, and taking most of our time: what is a qubit actually *made of*? A qubit is not an abstraction that happens to sit in a laboratory; it is a real physical system, held at a real temperature, driven by real pulses, and forgetting its state at a real rate. Understanding that physics explains where the mathematics comes from, why quantum computers are difficult to build, and why the machines being built today look the way they do.

The mathematics itself is deliberately kept to a minimum here. The two lectures that follow are devoted to it, including any proper construction.

Cheat Sheet

The six cardinal states, and what each measurement returns. Results are written in the ± 1 convention, in which the outcome 0 is reported as $+1$ and the outcome 1 as -1 ; *random* means the two results are equally likely. Each state has a definite result under exactly one of the three measurements, and the sign of that result is the sign of its Bloch axis.

State	Amplitudes	Bloch axis	Z meas.	X meas.	Y meas.
$ 0\rangle$	$(1, 0)$	$+\hat{z}$	$+1$	<i>random</i>	<i>random</i>
$ 1\rangle$	$(0, 1)$	$-\hat{z}$	-1	<i>random</i>	<i>random</i>
$ +\rangle$	$\frac{1}{\sqrt{2}}(1, 1)$	$+\hat{x}$	<i>random</i>	$+1$	<i>random</i>
$ -\rangle$	$\frac{1}{\sqrt{2}}(1, -1)$	$-\hat{x}$	<i>random</i>	-1	<i>random</i>
$ +i\rangle$	$\frac{1}{\sqrt{2}}(1, i)$	$+\hat{y}$	<i>random</i>	<i>random</i>	$+1$
$ -i\rangle$	$\frac{1}{\sqrt{2}}(1, -i)$	$-\hat{y}$	<i>random</i>	<i>random</i>	-1

Rules of Block 1.

Rule	Statement
state of a qubit	$ \psi\rangle = \alpha 0\rangle + \beta 1\rangle, \quad \alpha, \beta \in \mathbb{C}$
normalisation	$ \alpha ^2 + \beta ^2 = 1$
Born rule	$p(0) = \alpha ^2, \quad p(1) = \beta ^2$
global phase	$e^{i\gamma} \psi\rangle$ is the <i>same</i> physical state, for every γ
relative phase	$ 0\rangle + e^{i\phi} 1\rangle$: invisible to Z , visible to X and Y
orthogonality	perfectly distinguishable states are <i>antipodal</i> on the Bloch sphere
gates	rotations of the Bloch sphere; X, Z are π about $\hat{x}, \hat{z}$; H is π about $(\hat{x} + \hat{z})/\sqrt{2}$

Matrices, for reference (derived and used systematically in Lecture 3, Block 2).

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \quad Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \quad H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Physics of Block 2.

Quantity	Meaning
ω_{01}	qubit transition frequency; energy gap $\hbar\omega_{01}$ between $ 0\rangle$ and $ 1\rangle$
thermal condition	$\hbar\omega_{01} \gg k_B T$ is required for the qubit to rest in $ 0\rangle$
anharmonicity	$\omega_{12} \neq \omega_{01}$; without it a pulse leaks population out of the qubit
Ω	Rabi frequency; rotation rate on the Bloch sphere, set by the drive amplitude
pulse area	rotation angle $= \Omega t$; $\Omega t = \pi$ is an X gate, $\Omega t = \pi/2$ a quarter turn
drive phase ϕ	selects the rotation axis in the equatorial plane ($0 \rightarrow \hat{x}, \pi/2 \rightarrow \hat{y}$)
detuning Δ	tilts the rotation axis by $\arctan(\Delta/\Omega)$, rate becomes $\sqrt{\Omega^2 + \Delta^2}$
T_1	energy relaxation time, $ 1\rangle \rightarrow 0\rangle$; Bloch vector falls to the north pole
T_2	dephasing time; Bloch vector collapses onto the z axis. Always $T_2 \leq 2T_1$
figure of merit	gates before decoherence $\approx T_2/t_{\text{gate}}$
no-cloning	no device copies an unknown qubit state; linearity forbids it

Hardware glossary of Block 3.

Term	Meaning
transmon	superconducting qubit: an LC circuit whose inductor is a Josephson junction
Josephson junction	thin insulating barrier between superconductors; a lossless <i>nonlinear</i> inductor
dilution refrigerator	cryostat reaching 10–20 mK, needed by superconducting and spin qubits
Paul trap	radio-frequency electrode trap confining ions in ultra-high vacuum
motional mode	collective vibration of an ion chain, used as a bus for two-qubit gates
optical tweezer	tightly focused laser beam holding a single neutral atom; movable
Rydberg blockade	a highly excited atom shifts its neighbour's levels, preventing double excitation
dispersive readout	the qubit shifts a coupled resonator's frequency; a probe tone reads the shift
leakage	population escaping the chosen two levels into a third level
NISQ	noisy intermediate-scale quantum: devices too large to simulate, too noisy to correct
surface code	leading error-correcting code; a logical qubit from a two-dimensional patch of physical ones

1 Block 1: What the Experiments Force Us to Write Down

Learning goals:

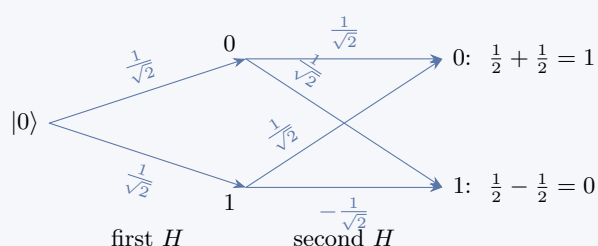
- read and write the ket notation for the six cardinal qubit states, and state the Born rule;
- explain why amplitudes rather than probabilities are the right bookkeeping, and where the squaring happens;
- tabulate the outcome statistics of the Z , X and Y measurements on those six states, and define what it means for two bases to be *mutually unbiased*;
- argue from the existence of a third measurement basis that the amplitudes must be complex numbers;
- distinguish a global phase, which is physically meaningless, from a relative phase, which is the resource every quantum algorithm spends;
- locate states on the Bloch sphere, recognise that distinguishable states are antipodal, and describe X , Z and H as rotations of it.

A short raison d'être

This block is short on purpose. Almost every statement in it is asserted rather than derived, because the derivations are the content of Lecture 3. What this block does instead is establish *why* the formalism has the shape it has. A student who meets “a qubit is a unit vector in $\mathbb{C}^2$ ” cold is entitled to ask three questions: why a vector, why complex, and why unit. The laboratory answers all three, and it is worth hearing the answers before the definitions arrive.

1.1 Where the Previous Lecture Left Us

The rule discovered in Lecture 1. *Multiply* amplitudes along a path, *add* them across paths with the same outcome, and *square* at the measurement gate, and only there.



Insert a measurement in the middle and the cancellation disappears: the two paths through the middle become two separate experiments, each with its own probabilities, and the fair-coin behaviour returns. *Interference survives only as long as nothing has recorded which path was taken.* We will meet that sentence twice more in Block 2, once as an optical experiment and once as the dominant error mechanism of real hardware.

1.2 Naming the States

The state that a Z measurement reports as 0 with certainty is written $|0\rangle$, and the one it reports as 1 with certainty is written $|1\rangle$. Lecture 3 explains why.

A general state carries an amplitude for each of the two outcomes,

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle, \quad (1)$$

and this is the object we will manipulate for the rest of the course. For now read (1) as a piece of bookkeeping: a state is whatever assigns an amplitude to each measurement outcome. That it is also a vector in a two-dimensional complex vector space, that the sum is a genuine vector-space sum and that $|0\rangle$ and $|1\rangle$ are an orthonormal basis of that space, is the subject of Lecture 3, Block 1.

Two features of (1) are worth flagging immediately. The qubit has exactly two basis states because it is a quantum *bit*. A system with three distinguishable outcomes for example would be a qutrit instead. And the coefficients are not restricted to being positive, or even real.

1.3 The Born Rule

The bridge from amplitudes to laboratory statistics is a single rule.

Born rule. Measuring $|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$ in the computational basis returns

$$p(0) = |\alpha|^2, \quad p(1) = |\beta|^2,$$

and leaves the qubit in the corresponding state $|0\rangle$ or $|1\rangle$. Consistency with $p(0) + p(1) = 1$ demands the *normalisation* condition

$$|\alpha|^2 + |\beta|^2 = 1. \quad (2)$$

This is a postulate

The Born rule is the one place where physics is injected into the formalism by hand. It is (a direct consequence of) an axiom of Quantum Physics, justified by the fact that a century of experiments agree with it and none disagree.

Example 1.1 (the state after a Hadamard). Symmetry fixes the state produced by H acting on $|0\rangle$ almost completely. It must give 0 and 1 with equal probability, so its two amplitudes must have equal modulus, and the most even-handed candidate is $c(|0\rangle + |1\rangle)$. Normalisation (2) then forces $2|c|^2 = 1$, so $c = 1/\sqrt{2}$ up to a phase, and we define

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad \text{with} \quad p(0) = p(1) = \left| \frac{1}{\sqrt{2}} \right|^2 = \frac{1}{2}.$$

The same argument admits a second solution, $|-\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$, which has identical statistics in the computational basis. Distinguishing $|+\rangle$ from $|-\rangle$ is exactly the business of the next section.

1.4 Three Measurements, Six States, One Experimental Fact

So far we have used only one measurement, the one that asks “are you $|0\rangle$ or $|1\rangle$?”. Nothing obliges us to ask only that question. A real device can be set to distinguish $|+\rangle$ from $|-\rangle$ just as sharply as it distinguishes $|0\rangle$ from $|1\rangle$ (at least, for the sake of the discussion here); in the

laboratory this is a different setting of the same apparatus, and in a circuit it is drawn by placing a Hadamard gate before an ordinary measurement. We call these the Z measurement and the X measurement, for reasons that Section 1.6 will make graphic.

There are an infinite amount of other settings, although only a third one is noteworthy. Alongside $|\pm\rangle$ there exists a pair of states, written $|\pm i\rangle$, which a third setting of the apparatus distinguishes perfectly, and which are even-handed with respect to *both* of the previous two settings. This is the Y measurement. Running all six states through all three settings produces the following table, which is the single most informative page of data in elementary quantum mechanics.

State	Amplitudes	$p(0)$ under Z	$p(0)$ under X	$p(0)$ under Y
$ 0\rangle$	$ 0\rangle$	1	$\frac{1}{2}$	$\frac{1}{2}$
$ 1\rangle$	$ 1\rangle$	0	$\frac{1}{2}$	$\frac{1}{2}$
$ +\rangle$	$(0\rangle + 1\rangle)/\sqrt{2}$	$\frac{1}{2}$	1	$\frac{1}{2}$
$ -\rangle$	$(0\rangle - 1\rangle)/\sqrt{2}$	$\frac{1}{2}$	0	$\frac{1}{2}$
$ +i\rangle$	$(0\rangle + i 1\rangle)/\sqrt{2}$	$\frac{1}{2}$	$\frac{1}{2}$	1
$ -i\rangle$	$(0\rangle - i 1\rangle)/\sqrt{2}$	$\frac{1}{2}$	$\frac{1}{2}$	0

Every one of the six states is *perfectly predictable* under exactly one of the three measurements, and *perfectly unpredictable*, an even coin toss, under the other two. There is no state anywhere in the table, and it turns out there is no state at all, that is simultaneously definite for two of these measurements.

Definition 1.2 (mutually unbiased bases). Two measurements are *mutually unbiased* if every state that gives a definite outcome for one gives a uniformly random outcome for the other. The Z , X and Y measurements of a qubit are pairwise mutually unbiased.

It is not a statement about the quality of our equipment: a better detector does not help. It is not a statement about our ignorance of some underlying value: the qubit is not secretly $+1$ for X while we look at Z . It is a statement that having a sharp value for one of these quantities is *incompatible* with having a sharp value for another. Bohr called this *complementarity*, and it is the physical content that the mathematics of Lecture 3 will encode. Block 2 shows the experiment that established it.

Why the amplitudes must be complex numbers. Suppose we tried to build the theory with real amplitudes only. Which real states are unbiased with respect to Z ? Normalisation gives $\alpha^2 + \beta^2 = 1$, and unbiasedness demands $\alpha^2 = \beta^2 = \frac{1}{2}$, so $\alpha = \pm 1/\sqrt{2}$ and $\beta = \pm 1/\sqrt{2}$. Discarding an overall sign, that leaves exactly two states: $|+\rangle$ and $|-\rangle$. In other words, *the only measurement unbiased with respect to Z that real amplitudes can supply is X itself*. A third mutually unbiased measurement simply does not fit. But the third measurement exists; the last two rows of the table are laboratory facts. Allowing $\beta = \pm i/\sqrt{2}$ produces states that are unbiased with respect to Z , because $|\pm i/\sqrt{2}|^2 = \frac{1}{2}$, and one checks that they are unbiased with respect to X as well. **The complex numbers are in quantum mechanics because the experiments put them there.** They are not a convenience of notation, and they cannot be removed without losing the Y measurement.

1.5 Which Phases Are Physical

We now have more mathematical descriptions than we have physically distinct states, and it is worth finding out how many are redundant.

Compare $|0\rangle$ with $-|0\rangle$. Under a Z measurement the first has $p(0) = |1|^2 = 1$ and the second has $p(0) = |-1|^2 = 1$: identical. That much is unsurprising, since the minus sign sits inside a modulus. The interesting check is whether some *other* measurement can see it. Writing $|0\rangle$ in terms of the X -basis states gives $|0\rangle = (|+\rangle + |-\rangle)/\sqrt{2}$, so an X measurement returns each outcome with probability $\frac{1}{2}$; and $-|0\rangle = (-|+\rangle - |-\rangle)/\sqrt{2}$ returns $|-1/\sqrt{2}|^2 = \frac{1}{2}$ for each as well. Identical again. The same happens for Y , and in fact for every measurement one can devise, because the overall factor of -1 multiplies every amplitude alike and is annihilated by every modulus.

Global phase is not physical. For any real γ , the states $|\psi\rangle$ and $e^{i\gamma}|\psi\rangle$ yield identical probabilities for every measurement whatsoever. Since measurement statistics exhaust what can be observed, the two *are* the same physical state, differently written.

Relative phase is physical. Compare instead

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} \quad \text{and} \quad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}.$$

Here the minus sign multiplies one amplitude and not the other, and no overall factor can remove it. The two states are indistinguishable under Z , both giving $\frac{1}{2}$, and perfectly distinguishable under X , giving 1 and 0. In general the phase ϕ in $(|0\rangle + e^{i\phi}|1\rangle)/\sqrt{2}$ is invisible to a Z measurement and completely determines the X and Y statistics.

1.6 The Bloch Sphere, as a Picture

Two real parameters survive after normalisation and the removal of the global phase, as the remark in Section 1.4 counted. A point on the surface of a sphere is also specified by two real parameters.

Discarding global phases, the states of a single qubit correspond *one-to-one* with the points on the surface of a unit sphere, the *Bloch sphere*. The six states of Section 1.4 sit at the six points where the coordinate axes pierce the surface, which is where the names Z , X and Y come from.

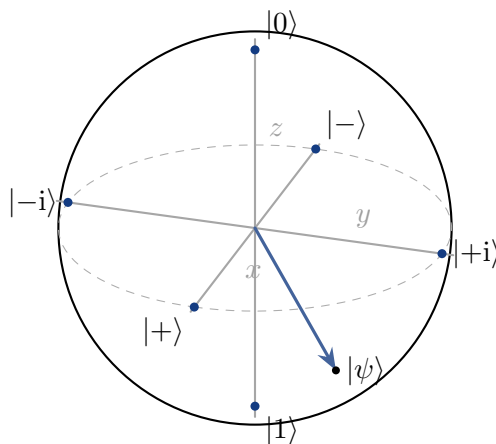


Figure 1: The Bloch sphere. Each point of the surface is one physical state of a qubit. The six states of Section 1.4 sit on the coordinate axes; antipodal pairs are the perfectly distinguishable pairs, and each such pair defines one possible measurement.

Three features of Figure 1 carry all the information we need from it.

Antipodal, not perpendicular. The perfectly distinguishable pairs, $|0\rangle$ and $|1\rangle$, $|+\rangle$ and $|-\rangle$, $|+i\rangle$ and $|-i\rangle$, sit at opposite ends of a diameter, and not at ninety degrees to one another. States that the mathematics of Lecture 3 will call *orthogonal* are antipodes on this picture. The reason is that the sphere is not the two-dimensional complex vector space itself but a real three-dimensional image of it, and the correspondence between the two doubles angles.

Every diameter is a measurement. There is nothing privileged about the three coordinate axes; any pair of antipodal points is a pair of perfectly distinguishable states, and therefore defines a measurement we could choose to perform. The habit of treating the Z axis as the default is a convention of quantum computing, inherited from the classical bit, and nothing more.

Superposition is not a mixture. A point on the equator is not a coin that has already landed and is being hidden from us. Such a coin would be described by the *centre* of the sphere (if at all!), not by a point on its surface; the interior is where classically random and decohered qubits live, and we return to it in Section 2.5.

Remark 1.3. The sphere is a picture of an abstract space, not of physical space; the x , y and z here are not directions in the laboratory. There is one important exception, and it is the subject of Section 2.2: for the spin of an electron or of a silver atom, the Bloch sphere very nearly *is* the space of directions in the room, and the three measurements are performed by pointing a magnet along three perpendicular axes.

The explicit parametrisation $|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\phi} \sin \frac{\theta}{2} |1\rangle$, the reason for the half-angle, and the conversion between amplitudes and Bloch coordinates are derived in Lecture 3, Block 2. We will not need them here.

1.7 Gates Are Rotations

A gate takes one state to another, hence one point of the sphere to another point. It cannot do so arbitrarily. It must send unit vectors to unit vectors, because probabilities must keep summing to one, so it maps the surface to itself. It must be reversible, because the evolution of an isolated quantum system runs equally well backwards, and because the previous lecture already needed the second Hadamard to undo the first. A transformation of a sphere onto itself that preserves distances and can be undone is a rotation. So gates are rotations (technically unitaries, that are slightly different from rotations for reasons important when considering at least two qubits), and the whole single-qubit gate set is the set of rotations of Figure 1.

The gates already met are the obvious ones.

- X is a half-turn about the x axis. It exchanges the poles, $|0\rangle \leftrightarrow |1\rangle$, which is why it is the quantum NOT gate, and it leaves $|+\rangle$ untouched, being a point on its own axis of rotation.
- Z is a half-turn about the z axis. It exchanges $|+\rangle \leftrightarrow |-\rangle$ and leaves the poles alone. It is to the X basis what X is to the Z basis.
- H is a half-turn about the tilted axis lying halfway between x and z . This is exactly why it maps $|0\rangle$ to $|+\rangle$ and back, and $|1\rangle$ to $|-\rangle$ and back, and why applying it twice returns every state to where it started: two half-turns about the same axis make a full turn.

Nothing restricts us to half-turns. A rotation by an arbitrary angle about an arbitrary axis is a legitimate gate, and Block 2 shows that on real hardware the angle is set by how long the control pulse lasts and the axis by the pulse's phase. This continuum of available gates has no classical counterpart: there is exactly one non-trivial gate on a classical bit, and there are infinitely many on a qubit.

Matrices, for reference only. Writing $|\psi\rangle$ as the column of its amplitudes (α, β) , the gates above act as

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \quad H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

They are collected here so that the notation is not unfamiliar when it returns.

Looking ahead. The three properties we have leaned on informally all have names, and all are established in the two lectures that follow. That states are vectors in $\mathbb{C}^2$ with an inner product, that $|0\rangle, |1\rangle$ are an orthonormal basis, and that the Bloch angles are θ and ϕ : Lecture 3, Blocks 1 and 2. That gates are the *unitary* operators, which is the precise version of “norm-preserving and reversible”: Lecture 4, Block 3, where it is also shown that unitaries preserve distinguishability, the algebraic shadow of their being rigid rotations. That a measurement is described by *projectors* and the Born rule extends to any basis: Lecture 4, Block 3. And X, Y, Z together with I turn out to be a basis for *all* operators on a qubit, the Pauli matrices of Lecture 4, Block 3.

2 Block 2: What a Qubit Is Made Of

Learning goals:

- explain the two conditions, a large energy gap and an anharmonic spectrum, that make a two-level subsystem of a real quantum system usable as a qubit, and say what *leakage* is;
- estimate from $\hbar\omega_{01} \gg k_B T$ why some platforms need a dilution refrigerator and others do not;
- describe the sequential Stern–Gerlach experiment and the three-polariser experiment, and identify both as realisations of the mutual-unbiasedness table of Block 1;
- read a Mach–Zehnder interferometer as the two-Hadamard circuit;
- explain how a resonant pulse rotates the Bloch vector, and relate pulse amplitude, duration, phase and detuning to the rotation rate, angle, axis and axis tilt;
- describe what physically happens during a readout, and why probabilities require many repetitions;
- define T_1 and T_2 , interpret dephasing as an unwanted measurement performed by the environment, and use T_2/t_{gate} as a figure of merit;
- prove the no-cloning theorem from linearity and state its consequences for error correction.

A short raison d'être

Block 1 treated the qubit as an abstraction that produces statistics. It is not. It is a superconducting circuit cooled to twenty thousandths of a degree above absolute zero, or a single ytterbium ion held in vacuum by radio-frequency electrodes, or one rubidium atom in the focus of a laser beam. Which of these you choose determines how fast your gates run, how long your information survives, which pairs of qubits can talk to each other, and how many qubits you can put on one machine. None of that is visible from the mathematics, and all of it decides what can actually be computed.

2.1 Why Two-Level Systems Exist At All

A quantum system that is bound, confined by a potential well of any shape, has a *discrete* spectrum of energies. This is the first thing anyone learns in a quantum mechanics course and it is the reason quantum computing is possible: it supplies, for free, systems with countably many distinguishable states rather than a continuum.

But a discrete spectrum is not yet a qubit. Almost every real system has far more than two levels, and we must be able to pick two and confine the dynamics to them. Two conditions do the work.

A large gap. Let $\hbar\omega_{01}$ be the energy separating the ground state, which we call $|0\rangle$, from the first excited state, which we call $|1\rangle$. If the system is in contact with an environment at temperature T , thermal fluctuations will excite it unless

$$\hbar\omega_{01} \gg k_B T. \tag{3}$$

When (3) holds, the system relaxes into $|0\rangle$ and stays there, which is exactly what we need: initialisation of a quantum computer to $|00\cdots 0\rangle$ is, on most platforms, nothing more elaborate than waiting.

An uneven ladder. Suppose the levels above $|1\rangle$ are spaced exactly like $|0\rangle$ and $|1\rangle$, so that $\omega_{12} = \omega_{01}$. Then any pulse tuned to drive $|0\rangle \rightarrow |1\rangle$ drives $|1\rangle \rightarrow |2\rangle$ just as efficiently, and the population climbs straight out of the two states we care about. This is what happens in a perfect harmonic oscillator, which is consequently useless as a qubit. What is needed is *anharmonicity*, $\omega_{12} \neq \omega_{01}$, so that a pulse resonant with the qubit transition is off-resonant with the next one up and leaves it alone. The residual population that escapes anyway is called *leakage*, and it is a genuine and distinctively quantum-hardware failure mode: the qubit has not flipped to the wrong value, it has left the computational space altogether.



Figure 2: A qubit is a chosen pair of levels inside a larger spectrum. The harmonic ladder on the left cannot supply one, because any pulse that drives $|0\rangle \rightarrow |1\rangle$ drives $|1\rangle \rightarrow |2\rangle$ equally well. Engineered anharmonicity or the natural irregularity of atomic spectra isolates the shaded pair.

Example 2.1 (the thermal condition decides the refrigerator). Condition (3) explains a great deal of what a quantum computing laboratory looks like. A superconducting qubit has $\omega_{01}/2\pi \approx 5$ GHz, and $\hbar\omega_{01} = k_B T$ at $T \approx 240$ mK. To have the qubit reliably in its ground state one needs a temperature well below that, which is why every superconducting machine sits at the bottom of a dilution refrigerator at 10–20 mK.

An ion, by contrast, stores its qubit in electronic levels whose separation is in the optical or microwave-to-optical range; for an optical qubit $\hbar\omega_{01}/k_B$ is of order 10^4 K. Equation (3) is then satisfied by an enormous margin in a room-temperature vacuum chamber, and no cryostat is needed. This does not make ions easier: their *motion* is thermally excited, motion is what mediates their two-qubit gates, and so it must be laser-cooled to near its own ground state instead. The difficulty is conserved; only its form changes.

2.2 Two Archetypes to Keep in Your Head

Before surveying the machines, it pays to have two concrete physical systems in mind, both of them older than quantum computing and both of them exactly a qubit.

Spin and the Stern–Gerlach experiment

An electron, or a silver atom carrying one unpaired electron, has an intrinsic angular momentum, its *spin*, and an associated magnetic moment. Send a beam of such atoms through an inhomogeneous magnetic field oriented along z and the beam is deflected. Classically one would expect a smear, since the moments should point in all directions. What Stern and Gerlach found

in 1922 was two spots. The component of the spin along the measurement axis takes exactly two values, and never anything in between.

That is a Z measurement, performed on a physical qubit, by pointing a magnet. Rotate the magnet to lie along x and it becomes an X measurement. And now the experiment can be run in sequence, which is where it becomes the most economical demonstration of everything in Section 1.4.

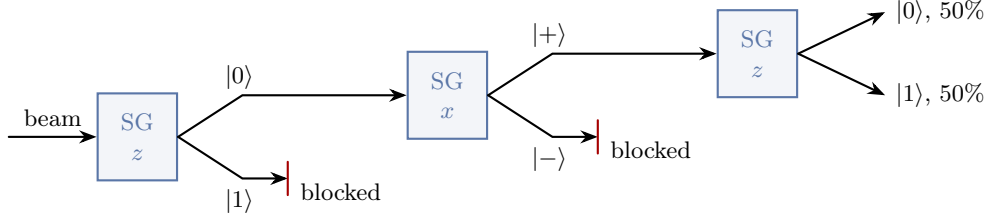


Figure 3: The sequential Stern–Gerlach experiment. After the first stage every atom is certainly $|0\rangle$. The second stage sorts by x and we keep only $|+\rangle$. A third stage sorting by z again finds the two outcomes equally likely: the x measurement has erased the z information that the first stage established.

After the first magnet, block the lower beam. Every surviving atom is now certainly $|0\rangle$: send it through a second z magnet and it goes up, every time. Now insert an x magnet in between and keep only its upper output, $|+\rangle$. The atoms reaching the final z magnet were *selected* to be $|0\rangle$ a moment ago, and yet they now split evenly. Measuring x has destroyed the value of z .

This is the mutual-unbiasedness table made of silver atoms, and it naively disposes of the obvious classical explanation (less naive, but more involved experiments exist, with the same conclusion). If the atom had merely been carrying a hidden pair of predetermined values, one for z and one for x , then filtering on x would have kept the z value intact and the final magnet would have sent everything up. It does not. There is no such pair of values to carry.

Polarisation and the interferometer

The polarisation of a photon is a second qubit, and it has the advantage that the corresponding experiment can be done on a kitchen table. Horizontal and vertical polarisation play the role of $|0\rangle$ and $|1\rangle$; the two diagonal polarisations, at $\pm 45^\circ$, are $|+\rangle$ and $|-\rangle$; the two circular polarisations are $|\pm i\rangle$. Cross two polarising filters at 90° and no light passes. Insert a third filter at 45° *between* them, and light reappears. Adding an obstacle increased the transmission, which is absurd unless the middle filter is not filtering but *measuring*, in a basis unbiased with respect to the other two. It is Figure 3 again, with sunglasses.

The second optical device to know is the Mach–Zehnder interferometer, because it is the two-Hadamard circuit of Section 1.1 built out of glass.

Here the qubit is not a property of the photon at all but the *path* it takes: $|0\rangle$ means the lower arm, $|1\rangle$ the upper. A balanced beam splitter sends an incoming photon into an equal superposition of the two arms, which is precisely the action of a Hadamard. A piece of glass in one arm delays that branch and imprints a relative phase ϕ , which is a rotation about the z axis. The second beam splitter recombines. With $\phi = 0$ every photon arrives at the same detector, in perfect analogy with the two-Hadamard circuit, and sweeping ϕ moves them sinusoidally from one detector to the other.

Now put a detector in one arm to find out which way the photon went. The fringes vanish and each output detector fires half the time. This is the measure-in-the-middle circuit, and it is the second appearance of the sentence from Section 1.1: interference survives only as long as

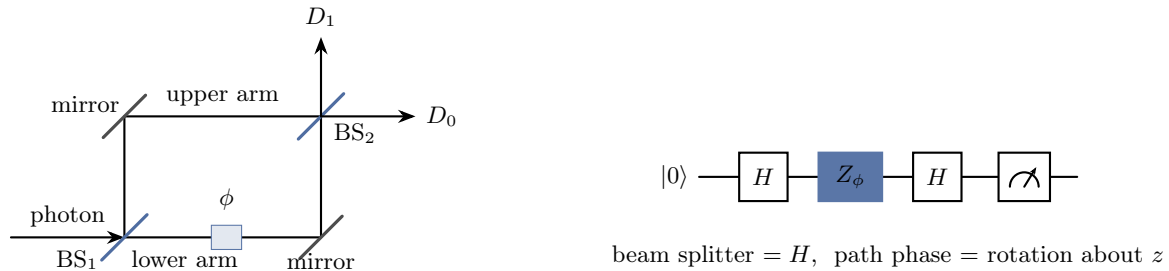


Figure 4: A Mach–Zehnder interferometer and the circuit it implements. The qubit is *which arm the photon is in*. The first beam splitter is a Hadamard, the phase plate is a z rotation by ϕ , the second beam splitter is a second Hadamard. With $\phi = 0$ every photon arrives at D_0 , which is the two-Hadamard result of Lecture 1.

nothing has recorded which path was taken. Note that the recording is what matters, not the disturbance: the interference is lost even when the which-path information is never looked at, as long as it exists somewhere.

One qubit, three incarnations. A superconducting circuit’s two lowest energy levels, a silver atom’s spin along z , and a photon’s choice of arm are three utterly different physical things. They obey the same table of statistics, they are rotated by the same gates, and they are described by the same two complex amplitudes.

2.3 How a Gate Physically Happens

Block 1 established that a gate is a rotation of the Bloch sphere. We can now say what performs the rotation.

An idle qubit is not idle

Because $|0\rangle$ and $|1\rangle$ have different energies, the relative phase between their amplitudes advances by itself, at the rate ω_{01} . On the Bloch sphere this is a steady rotation about the z axis: a qubit prepared at $|+\rangle$ and left alone circles the equator some five billion times a second. The poles are unaffected, since they lie on the rotation axis, but every superposition is in constant motion.

Rather than track that motion, everyone works in the *rotating frame*: describe the state relative to a clock running at ω_{01} , in which the free precession is subtracted away and an undisturbed qubit simply stands still. Everything that follows is stated in this frame. One consequence is worth noting immediately. If the frame is only a bookkeeping convention, then changing its phase is free, and a rotation about z can be implemented by *redefining the phase of every subsequent pulse* rather than by doing anything to the qubit at all. Such *virtual Z gates* are standard on superconducting hardware: they take zero time and introduce essentially zero error, which is why compilers work hard to push z rotations into that form.

Driving

To rotate about any other axis we must intervene, by applying an oscillating field at, or near, the qubit’s own frequency: a microwave pulse down a coaxial line for a transmon, a laser pulse for an ion or a neutral atom, an oscillating magnetic field for a spin. In the rotating frame a resonant drive produces a steady rotation about an axis lying *in the equatorial plane*, and three knobs control it.

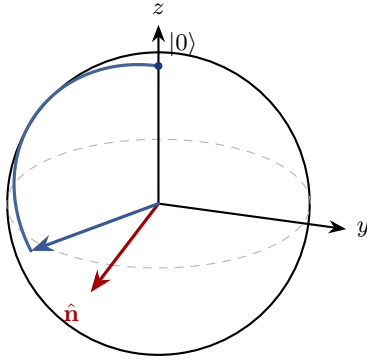
- The **amplitude** of the drive sets the rotation rate, called the *Rabi frequency* Ω . Turn the power up and the Bloch vector turns faster.
- The **phase** of the drive selects which equatorial axis is the rotation axis. Phase 0 gives a rotation about $\hat{x}$, phase $\pi/2$ a rotation about $\hat{y}$, and intermediate phases everything in between. This is why a single physical control can produce a two-parameter family of gates.
- The **duration** t , together with Ω , fixes how far round the state travels. The rotation angle is the *pulse area*

$$\Theta = \Omega t. \quad (4)$$

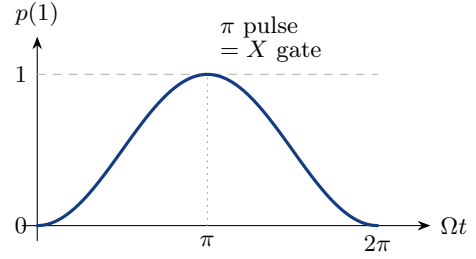
Equation (4) is the practical heart of quantum control. A π pulse, $\Omega t = \pi$, is a half-turn: applied about $\hat{x}$ to a qubit in $|0\rangle$ it produces $|1\rangle$, and it *is* the X gate. A $\pi/2$ pulse is a quarter turn, taking $|0\rangle$ to the equator, and two of them with suitably chosen phases build a Hadamard. Starting from $|0\rangle$ and measuring in the computational basis, the probability of finding $|1\rangle$ is

$$p(1) = \sin^2\left(\frac{\Omega t}{2}\right), \quad (5)$$

the *Rabi oscillation*. Sweeping the pulse duration and plotting the result is the first measurement made on any new device, because the position of the first maximum calibrates the π pulse and hence every gate built from it.



(a) a drive of phase 0 rotates about $\hat{n} = \hat{x}$ at rate Ω



(b) Rabi oscillation $p(1) = \sin^2(\Omega t/2)$

Figure 5: Driving a qubit. A resonant pulse rotates the Bloch vector about an equatorial axis chosen by the drive's phase, at a rate set by its amplitude. Sweeping the duration traces out the Rabi oscillation, whose first maximum calibrates the π pulse.

Detuning

Suppose the drive frequency misses the qubit frequency by $\Delta = \omega_{\text{drive}} - \omega_{01}$. Two things change. The rotation axis tilts out of the equatorial plane, towards $\hat{z}$, by an angle $\arctan(\Delta/\Omega)$; and the rotation speeds up, to $\sqrt{\Omega^2 + \Delta^2}$. The consequence is that a badly detuned pulse can never fully invert the qubit: the Bloch vector precesses about a tilted axis and misses the south pole, so $p(1)$ oscillates between 0 and $\Omega^2/(\Omega^2 + \Delta^2)$ rather than reaching 1.

Detuning is therefore an error mechanism, since qubit frequencies drift and are never known perfectly. It is also, read the other way round, a *resource*: it is precisely because a neighbouring qubit is detuned from the pulse that the pulse does not act on it. Frequency is how most platforms address individual qubits, and the same argument, applied to the $|1\rangle \rightarrow |2\rangle$ transition, is how anharmonicity suppresses leakage.

For readers with a physics background

In the rotating frame the driven qubit is governed by the time-independent Hamiltonian

$$H = -\frac{\hbar}{2} \left[\Delta Z + \Omega (\cos \phi X + \sin \phi Y) \right],$$

where ϕ is the drive phase, and the state obeys the Schrödinger equation $i\hbar \partial_t |\psi\rangle = H |\psi\rangle$. Its solution is a rotation of the Bloch vector about the axis

$$\hat{\mathbf{n}} \propto (\Omega \cos \phi, \Omega \sin \phi, \Delta), \quad \text{at rate} \quad \Omega_{\text{eff}} = \sqrt{\Omega^2 + \Delta^2},$$

which is every statement of this section at once. Writing the solution as $U = \exp(-iHt/\hbar)$ requires the exponential of an operator, defined through its diagonal representation in Lecture 4, Block 4; the worked example there, $e^{-i\theta X} = \cos \theta I - i \sin \theta X$, is the $\Delta = 0$, $\phi = 0$ case of this formula.

Two-qubit gates need an interaction

Everything above concerns one qubit at a time, and no sequence of such pulses will ever entangle anything. A pulse applied to one qubit acts on that qubit alone, so a circuit built only from them leaves each qubit with a state of its own, however elaborate the schedule; for the same reason that no amount of independent stirring makes two teacups correlated. To build a gate whose effect on one qubit *depends* on the state of another, the two must physically interact.

This is a genuine physical requirement and not a mathematical inconvenience, and it is the hardest one on the list. A qubit must be strongly coupled to its partner when a gate is being applied and strongly isolated from everything else, including that same partner, at all other times. Those two demands pull in opposite directions, which is why two-qubit gates are, on every platform without exception, slower and roughly an order of magnitude less accurate than single-qubit gates, and why the entangling gate is the figure of merit by which a machine is usually judged.

What supplies the interaction is the single largest engineering difference between the technologies, and it is where the platforms of Block 3 genuinely part company: each one is described there together with the coupling mechanism it uses. The mathematics of two-qubit states, the tensor product, is Lecture 4, Block 4.

2.4 What a Measurement Physically Is

In Block 1 measurement was a rule: square the amplitudes, get a probability. In the laboratory it is a piece of apparatus, and the apparatus is instructive.

A measurement is a deliberate, strong coupling of the qubit to a large system whose states are macroscopically distinct, arranged so that the qubit's two states drive that system to two different, and irreversibly distinguishable, outcomes. Two examples cover most hardware.

Fluorescence readout, used by ions and neutral atoms, shines a laser tuned so that one qubit state scatters photons vigorously while the other is transparent to it. A camera or photomultiplier then sees the atom as bright or dark; a single ion may scatter of order 10^6 photons in a millisecond, so collecting even a small fraction settles the question beyond doubt.

Dispersive readout, used by superconducting qubits, couples the qubit to a microwave resonator detuned from it. The qubit's state pulls the resonator's frequency by a small amount, and a weak probe tone reflected off the resonator comes back with a phase that differs measurably between the two cases.

Both are slow by the standards of the gates: readout typically takes hundreds of nanoseconds

to milliseconds, against tens of nanoseconds to microseconds for a gate. Both are imperfect, misreporting the state with a probability that is usually the worst error rate in the machine. And both are irreversible, which is the point: the amplification into a macroscopic record is what makes a measurement a measurement rather than just another unitary rotation.

One run gives one bit. A single execution of a circuit yields a single outcome, 0 or 1, and tells you almost nothing. Every probability written in Block 1, and every histogram produced by a real device, is an estimate obtained by preparing the same state and measuring it again, thousands of times over. A run of N repetitions determines a probability to about $1/\sqrt{N}$, so distinguishing $p = 0.50$ from $p = 0.52$ takes of order 10^4 shots. This is why quantum computers are operated as sampling machines, and why an algorithm whose answer is encoded in a tiny difference of probabilities is, for practical purposes, no algorithm at all.

2.5 Decoherence: the Real Enemy

No qubit is isolated. It sits in a substrate that vibrates, near defects that fluctuate, in a magnetic field that drifts, wired to control lines that carry noise. The environment therefore interacts with the qubit constantly and, in effect, measures it. Two timescales quantify the damage.

T_1 , energy relaxation. The excited state can give its energy up to the environment and decay, $|1\rangle \rightarrow |0\rangle$, with a characteristic time T_1 . On the Bloch sphere the vector drifts towards the north pole. This is spontaneous emission, or its solid-state equivalent, and it destroys the populations themselves.

T_2 , dephasing. Even with no energy exchanged, the relative phase between the two amplitudes is randomised: a fluctuating environment makes ω_{01} jitter, so the state accumulates an unpredictable extra phase. On the Bloch sphere the equatorial component of the vector shrinks while its z component is untouched, and the vector collapses onto the z axis. Since relaxation also destroys phase coherence, the two rates are not independent, and one always has

$$T_2 \leq 2T_1. \quad (6)$$

Dephasing is the environment measuring your qubit. A stray degree of freedom that becomes correlated with whether the qubit is $|0\rangle$ or $|1\rangle$ has recorded which-path information, exactly as the detector in the interferometer arm of Section 2.2 did, and exactly as the measurement gate inserted between the two Hadamards of Section 1.1 did. The interference on which the computation depends then disappears. This is the third and final appearance of that sentence, and the three are not analogies but one phenomenon: *decoherence is unwanted measurement, performed continuously, by an environment we cannot control.* The only difference from a deliberate readout is that nobody reads the result.

The figure of merit that follows is simple and unforgiving. If a gate takes t_{gate} and coherence lasts T_2 , then the number of gates that can be executed before the information is gone is roughly

$$N_{\text{gates}} \approx \frac{T_2}{t_{\text{gate}}}. \quad (7)$$

Neither factor alone means anything: ions have coherence times a hundred thousand times longer than transmons and gates a thousand times slower, and the comparison that matters is the ratio. Block 3 tabulates it across the platforms.

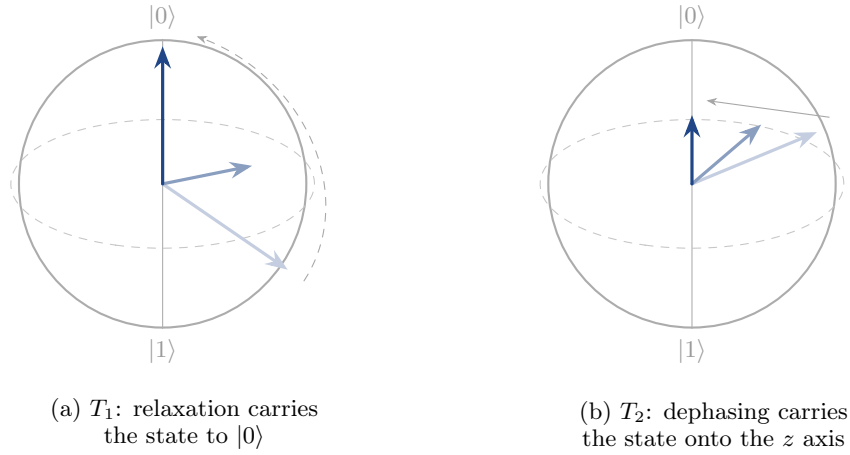


Figure 6: The two elementary decoherence processes, drawn as motions of the Bloch vector. Relaxation carries any state to $|0\rangle$, which is still a point of the surface. Dephasing shrinks the equatorial component while leaving the z component alone, so the vector ends on the axis and, in general, strictly *inside* the sphere, where it is no longer of the form $\alpha |0\rangle + \beta |1\rangle$ at all.

Remark 2.2 (where the interior of the Bloch sphere comes in). A decohered qubit is no longer described by two amplitudes at all. Its Bloch vector is shorter than one, and it lies strictly inside the sphere; the centre represents a qubit about which nothing whatsoever is known. Such states are called *mixed*.

2.6 Why You Cannot Copy a Qubit to Fight Noise

Faced with an unreliable classical bit, the obvious defence is redundancy: store three copies and take a majority vote. It is worth understanding why this elementary idea is unavailable here, because the answer shapes everything in Section 3.4.

Theorem 2.3 (no-cloning). *There is no quantum gate that copies an arbitrary unknown qubit state.*

Heuristic “proof”. Suppose some device C took a qubit in an unknown state together with a blank qubit and produced two copies. On the two states we care most about it would act as

$$C |0\rangle |0\rangle = |0\rangle |0\rangle, \quad C |1\rangle |0\rangle = |1\rangle |1\rangle.$$

But gates are linear: their action on a superposition is fixed by their action on the pieces. Applying C to $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$ therefore gives

$$C |+\rangle |0\rangle = \frac{1}{\sqrt{2}} (C |0\rangle |0\rangle + C |1\rangle |0\rangle) = \frac{|0\rangle |0\rangle + |1\rangle |1\rangle}{\sqrt{2}}.$$

A genuine copier would have had to produce $|+\rangle |+\rangle$, which expands to $\frac{1}{2}(|0\rangle |0\rangle + |0\rangle |1\rangle + |1\rangle |0\rangle + |1\rangle |1\rangle)$ and has four terms rather than two. The two results differ, so C does not copy $|+\rangle$. A device that copies the basis states cannot copy their superpositions. $\square$

Remark 2.4. The notation $|0\rangle |0\rangle$ for “both qubits are in $|0\rangle$ ” is used here informally; it is the tensor product, defined properly in Lecture 4, Block 4, where the state $(|0\rangle |0\rangle + |1\rangle |1\rangle)/\sqrt{2}$ appears again as the canonical *entangled* state. The proof above needs nothing beyond linearity and the fact that $|+\rangle$ is a superposition, which is why it can be given now.

The consequences are pervasive. There is no majority vote, because there are no copies to vote. There is no amplifier for a quantum signal, which is why quantum networks cannot use repeaters of the classical kind. There is no way to learn both α and β from a single qubit, since one could otherwise copy it many times and measure the copies in different bases; a single qubit yields a single bit, as Section 2.4 insisted. And error correction, when we come to it, cannot work by inspecting the data: it must extract information about *whether an error occurred* while learning nothing about the state itself. That is a surprising requirement, it is satisfiable, and Section 3.4 says how.

Looking ahead. The physical statements of this block become mathematical ones in the lectures that follow. That a gate is norm-preserving and reversible becomes *unitarity*, $U^\dagger U = I$, with reversibility and the preservation of distinguishability as theorems. That a measurement returns one outcome and leaves the qubit in the corresponding state becomes the algebra of *projectors*. The rotation generated by a driving Hamiltonian becomes the *operator exponential* $e^{-i\theta X} = \cos \theta I - i \sin \theta X$. And the two-qubit notation used informally in the no-cloning proof becomes the *tensor product*, along with the definition of entanglement.

3 Block 3: How Quantum Computers Are Actually Built

Learning goals:

- state the DiVincenzo criteria and use them as a checklist against which any proposed technology can be read;
- for each of the main platforms, say what the two levels physically are, how the qubit is initialised, driven, coupled and read out;
- compare platforms on gate speed, coherence, connectivity and scale, and explain why no single number decides the comparison;
- explain the gap between physical and logical error rates, and why error correction is not optional;
- describe in outline how a code protects information without ever measuring it, and reconcile this with the no-cloning theorem.

3.1 The DiVincenzo Checklist

In 2000 David DiVincenzo wrote down the conditions any physical system must satisfy to serve as a quantum computer. They remain the standard way to assess a proposal, and every platform below should be read against them.

1. **A scalable system of well-characterised qubits.** Well-characterised means that the two levels are known, that the rest of the spectrum is known well enough to avoid leakage, and that the couplings are understood. Scalable means that adding the thousandth qubit is not a fundamentally different problem from adding the tenth.
2. **Initialisation** to a known state such as $|00 \cdots 0\rangle$. By Section 2.1 this is often just cooling and waiting, though fast active reset matters once the machine is running deep circuits.
3. **Coherence times long compared with gate times**, the ratio (7). Note that this is a ratio and not a duration.
4. **A universal set of gates.** Arbitrary single-qubit rotations plus essentially any entangling two-qubit gate suffice to approximate any computation.
5. **Qubit-specific measurement**, and preferably measurement that can be performed mid-circuit on selected qubits without disturbing the rest, since error correction depends on exactly that.

Two further criteria govern quantum *communication*: the ability to convert stationary qubits into flying ones (photons), and to transmit those faithfully between locations.

The interesting thing about the list is that no platform is uniformly good at all of it. Each technology is excellent at some criteria and awkward at others, and the choices are not independent: what makes a qubit easy to control also makes it easy for the environment to disturb.

3.2 The Platforms

Superconducting circuits

Take an electrical circuit, make it out of a superconductor so that resistance and hence dissipation vanish, and cool it until thermal noise is negligible. An ordinary LC resonator is then a quantum harmonic oscillator, with the evenly spaced ladder of Figure 2, and by Section 2.1

it is useless. The trick that makes the field work is to replace the inductor with a *Josephson junction*, a thin insulating barrier between two superconductors, which behaves as a lossless *nonlinear* inductor. The ladder becomes anharmonic, and the lowest two rungs are a qubit. The standard design is the *transmon*.

The two levels are the two lowest states of this artificial atom, separated by $\omega_{01}/2\pi \approx 4\text{--}6$ GHz. Initialisation is thermal, at 10–20 mK in a dilution refrigerator, optionally accelerated by active reset. Gates are microwave pulses delivered by coaxial line, tens of nanoseconds long, with z rotations done virtually as in Section 2.3. Two-qubit gates use the electrical coupling between neighbouring circuits on the chip, either a shared capacitance or an intermediary bus resonator, whose strength may be fixed by the layout or made tunable by a coupler element placed between the two qubits. Readout is dispersive.

The attraction is that these are lithographically fabricated circuits: the design is engineered rather than given by nature, the gates are the fastest of any platform, and the manufacturing draws on the semiconductor industry. The difficulties are that qubits are confined to a two-dimensional chip with only nearest-neighbour coupling, that every qubit needs its own microwave lines running into a refrigerator with finite cooling power and finite room for cabling, that no two fabricated qubits come out quite identical, and that coherence times, while enormously improved, remain among the shortest in the field.

Trapped ions

Strip an electron from an atom and the resulting ion can be held in vacuum by oscillating electric fields, in a *Paul trap*, essentially forever. Laser cooling brings it nearly to rest. The qubit is a pair of the ion’s own electronic states: either two hyperfine levels of the ground state, separated by a few GHz and extraordinarily stable, or a ground state and a long-lived metastable state separated by an optical frequency.

Because the levels are supplied by nature, every ion of a given species is *identical*, which removes an entire class of calibration problems. Initialisation is by optical pumping, single-qubit gates are laser or microwave pulses, and readout is fluorescence: bright or dark, with excellent fidelity. Ions in a chain repel one another, so they vibrate collectively; a laser pulse can couple a chosen ion’s internal state to that shared vibration, and a second ion can be coupled to the same mode. The motion acts as a bus, and any pair of ions in the chain can be entangled directly, giving effectively *all-to-all connectivity*.

Ion machines hold the highest demonstrated gate and readout fidelities, and coherence times of seconds or better. The costs are speed, since two-qubit gates take tens to hundreds of microseconds, and scale: a single chain becomes unwieldy beyond a few tens of ions, because the vibrational modes crowd together. The leading answer is the *QCCD* architecture, in which ions are physically shuttled between separate trap zones on a chip, which trades the connectivity advantage back for engineering complexity.

Neutral atoms and the Rydberg blockade

A neutral atom, unlike an ion, carries no charge and so barely interacts with its neighbours or with stray fields, which is excellent for coherence and useless for gates. Individual atoms are captured in *optical tweezers*, tightly focused laser beams, and arranged into arrays of hundreds to thousands of sites; the qubit lives in hyperfine or optical-clock levels, as for an ion.

The interaction is switched on when required. Excite an atom to a *Rydberg* state, with its outer electron promoted to a very high orbit, and it acquires an enormous electric dipole moment that shifts the levels of any nearby atom. In the *Rydberg blockade*, two atoms close enough together cannot both be excited, and that conditional behaviour is exactly what a

two-qubit gate needs. Turning the excitation lasers off restores the atoms to near-isolation.

Two features are distinctive. Arrays are large: thousands of trap sites are routine, since a lattice can be made of as many tweezers as the laser power allows. And the tweezers can be *moved* during a computation, physically carrying atoms next to whichever partners they must interact with, so the connectivity graph is reconfigurable rather than fixed by fabrication. The characteristic difficulties are that atoms are occasionally lost from their traps and must be detected and reloaded, that gate fidelities have historically trailed ions and superconductors, and that fluorescence readout has to be engineered carefully to be non-destructive and to work mid-circuit.

Photonics

A photon can carry a qubit in its polarisation, in which of two paths it occupies (*dual rail*), or in which of two time slots it arrives. Photons do not decohere in any ordinary sense: a photon in free space has no environment to speak of, and there is no T_1 or T_2 to quote. Single-qubit gates are passive optics, wave plates and beam splitters and phase shifters, and they are essentially perfect. Section 2.2 was already a photonic quantum circuit.

The difficulty is exactly the mirror image of everyone else's. Photons do not interact with one another, so there is no direct two-qubit gate. The standard solutions are to make the gate *probabilistic*, using interference followed by measurement, and to correct the failures by feed-forward; or to prepare a large entangled resource state in advance and compute by measuring it, the measurement-based model; or to abandon single photons for squeezed states of light and continuous-variable encodings. All of these demand extremely low optical loss, fast switching, and single-photon detectors, which do need cryogenics even though the optics do not. The compensating advantages are room temperature operation for much of the apparatus, natural compatibility with optical fibre and hence with networking, and a manufacturing path through integrated photonics.

Semiconductor spin qubits

Confine a single electron in a *quantum dot*, a small region of a silicon device defined by voltages on metal gates, and apply a magnetic field. The electron's spin then has two levels split by the Zeeman energy, and that is the qubit. Silicon is a favourable host because the dominant isotope has no nuclear spin, and isotopic purification removes most of the remaining magnetic noise.

Gates are driven by oscillating magnetic or electric fields, and two-qubit gates use the exchange interaction: lower the barrier between two dots so that the electron wavefunctions overlap, and the spins interact strongly and very quickly. Readout converts spin into charge, by arranging for one spin state and not the other to be able to tunnel out of the dot, and then detecting the charge.

The case for spin qubits is size and manufacturing. A quantum dot is tens of nanometres across, some hundred thousand times smaller in area than a transmon, and the devices are made in the same foundries and with the same processes as classical chips. Some designs operate above one kelvin, which relieves the refrigeration problem considerably. Against that, device-to-device uniformity remains hard, charge noise limits coherence, and the number of qubits demonstrated in a single working processor is still small compared with the leading platforms.

Topological qubits

Every technology above fights noise after the fact. The topological approach proposes to prevent it: encode the information *non-locally*, in a property shared between two distant parts of a system, so that no local disturbance can either read it or corrupt it. The most pursued route

uses Majorana zero modes at the ends of a semiconductor nanowire coupled to a superconductor, and the protection would be inherited from the physics of the material rather than implemented in software.

If it worked at scale it would be a decisive advantage, because it attacks the problem of Section 3.4 at the root. While the maths behind is sound, their observations are controversial. Establishing that the required quasiparticles are present and behave as predicted has itself been difficult and contested, and there are as yet no topological processors running circuits.

NV center and NMR

Two older platforms deserve a mention. *Nitrogen-vacancy centres* in diamond are atom-like defects with spin qubits that work at room temperature and couple naturally to light; they are unlikely to scale into processors but are excellent quantum sensors and promising network nodes. *Nuclear magnetic resonance* on molecules in solution ran the first demonstrations of quantum algorithms in the late 1990s, and taught the field most of what it knows about pulse sequences, but it does not scale at all: the signal from a genuinely quantum state falls off exponentially with the number of spins.

3.3 Comparing the Platforms

Platform	Temp.	2Q gate	Coherence	T_2/t_{2Q}	Connectivity	Main obstacle
Superconducting	10–20 mK	10^2 ns	$\sim 10^2 \mu\text{s}$	$\sim 10^3$	nearest neighbour, fixed	cryogenic wiring, crosstalk, device spread
Trapped ions	room temp.	$10\text{--}10^2 \mu\text{s}$	$1\text{--}10^2$ s	$\sim 10^5$	all-to-all in a chain	gate speed; scaling past one chain
Neutral atoms	room temp.	$\sim 1 \mu\text{s}$	$1\text{--}10$ s	$\sim 10^5$	reconfigurable, atoms move	atom loss; mid-circuit readout
Photonics	optics warm	probabilistic	loss-limited	—	set by the optical network	no direct 2Q gate; photon loss
Spin in silicon	0.1–1 K	$10\text{--}10^2$ ns	$10 \mu\text{s}\text{--}1$ ms	$\sim 10^3$	nearest neighbour, fixed	uniformity, charge noise, wiring
Topological	mK	—	—	—	—	demonstrating the qubit at all

These are order-of-magnitude figures for typical devices around 2025–26, gathered to show the *shape* of the trade-offs rather than to rank anybody. Every entry is a moving target, best results run ahead of typical ones by an order of magnitude or more, and the numbers quoted by different groups are often not measured the same way. Consult current literature and vendor roadmaps before relying on any of them.

In brief: *No single number decides anything*: superconducting qubits lose on coherence by five orders of magnitude and win it all back on gate speed, so the ratio that matters is comparable. *Connectivity is as important as fidelity*: a circuit written for all-to-all hardware must be compiled onto a nearest-neighbour chip by inserting swap operations, which can multiply the gate count several-fold and consume the fidelity advantage the chip appeared to have. And *the obstacles are not the same kind of problem*: some are physics, some are engineering, and it is not obvious in advance which is harder.

3.4 From Physical to Logical Qubits

Every machine described above makes errors at a rate that is, by the standards of computing, catastrophic. A good two-qubit gate today fails with probability of order 10^{-3} , the best of

them somewhat better. An algorithm that runs a billion gates and expects a meaningful answer needs a per-gate error below roughly 10^{-9} , six orders of magnitude beyond what any hardware provides, and no plausible improvement in materials will close that gap on its own.

It is tempting to complete the comparison by quoting an error rate per operation for a classical processor, and figures around 10^{-18} circulate widely for the purpose. They should be treated with caution: classical logic gates are reliable enough that their failure rate per operation is not a quantity anyone routinely measures, and such numbers are back-of-the-envelope estimates rather than measurements. What *is* measured, at scale, is less flattering than the folklore. In a two-and-a-half-year study of Google’s server fleet, Schroeder, Pinheiro and Weber found DRAM error rates “orders of magnitude higher than previously reported”, between 25 000 and 70 000 failures per billion device-hours per Mbit, with more than 8% of memory modules affected per year and about a third of all machines seeing at least one memory error annually.¹

The reason no user ever notices is that the hardware corrects itself: server memory is protected by error-correcting codes, which repair single-bit faults silently and report the rest. This is worth saying plainly, because it removes the impression that what follows is a quantum peculiarity. *Error correction is what computing does whenever its hardware is not good enough*, and classical machines adopted it decades ago in exactly those places, memory and storage and communication links, where the raw physics was not reliable enough to trust. Quantum computers need the same remedy, only far more of it, and with one extra difficulty that has no classical counterpart: by Section 2.6, the redundancy cannot be built out of copies.

It has to be closed in software, by *quantum error correction*, and the reason this is possible at all is the threshold theorem: provided the physical error rate is below some critical value, encoding one logical qubit into many physical ones suppresses the logical error rate exponentially in the size of the code. Below threshold, accuracy becomes a matter of spending more qubits. Above it, no amount of redundancy helps.

The leading candidate is the *surface code**, which encodes a logical qubit in a two-dimensional patch of physical qubits with only nearest-neighbour interactions, a requirement chosen deliberately to suit chip-based hardware. Its threshold is around one per cent for the standard noise model, which is why the 10^{-3} gate errors of Section 3.3 are the milestone everyone has been chasing: they are just below it. The price is the code’s size. Reaching logical error rates useful for large algorithms takes of order 10^2 to 10^3 physical qubits per logical qubit, and more for the most demanding applications, which is why estimates for cryptographically relevant machines run to millions of physical qubits.

How a code obeys the no-cloning theorem. Section 2.6 ruled out copying the state, so error correction cannot work by storing duplicates and comparing them. What it does instead is measure *parities*. A circuit repeatedly asks questions of the form “do these two neighbouring qubits agree?”, which can be answered without learning what either of them is: the answer is the same whether the pair is $|00\rangle$ or $|11\rangle$. The collection of answers, the *syndrome*, identifies where an error occurred without revealing, and therefore without destroying, the encoded information. Redundancy is spread across correlations rather than across copies, which is exactly the loophole the no-cloning theorem leaves open.

Where does this leave the machines of today? They are in what is called the *NISQ* regime, noisy intermediate-scale quantum: large enough that simulating them classically is expensive, too noisy to run corrected circuits of useful depth. Error correction has moved in the last few years from theory to experiment, with small codes demonstrated on several platforms and the

¹B. Schroeder, E. Pinheiro and W.-D. Weber, “DRAM Errors in the Wild: A Large-Scale Field Study”, *SIGMETRICS* 2009. The same study found the errors to be dominated by hard failures rather than by the cosmic-ray soft errors that earlier work had assumed.

key signature, a logical error rate that *falls* as the code is made larger, observed. That is the crossing of a genuine threshold in the field's history, and it is also a long way from a fault-tolerant machine running deep algorithms. Any statement about where the frontier lies dates quickly, this one included; check the current roadmaps.

The abstraction holds

A transmon is a superconducting circuit patterned in aluminium and held at fifteen millikelvin. An ion is a single charged atom levitated in vacuum. An atom in a tweezer is held by light; a photon is not held at all; a spin qubit is one electron under a metal gate. These have almost nothing in common physically.

Yet every one of them presents the same interface to whoever writes the algorithm. A state is described by amplitudes, one per outcome, whose squared moduli are the probabilities and which sum to one. Operations are reversible and preserve that sum. A measurement returns one outcome and leaves the system in the corresponding state. Putting two of them side by side gives a system whose description is the product of the descriptions, and which can be entangled.

That interface is exactly what the next two lectures construct:

- states as unit vectors in $\mathbb{C}^2$, with the Bloch parametrisation of the picture used throughout this lecture;
- gates as *unitary* operators, the precise form of “rotation”;
- measurements as *projectors*, with the Born rule for a general basis;
- registers, two-qubit gates and entanglement as the *tensor product*;
- mixed states and the interior of the Bloch sphere, needed to describe the decoherence of Section 2.5.

The reason to build that mathematics once, abstractly, is now clear: it is the only part of the subject that does not change when the hardware does.